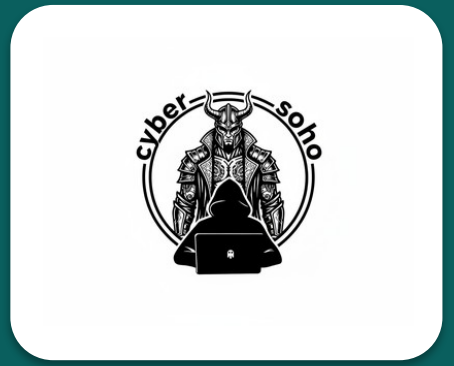
DIG 123 · AGILE METHODOLOGIES FOR INNOVATION · CLASS 7

Agile in the digital business: products, services and innovation frameworks

Your product leaves the classroom: who builds it, who uses it, who pays for it, and how cheaply it can change

Monday, November 2, 2026 · Sprint 2, second week

Lab 04 — Flow & Digital Business Brief opens in today's studio



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7 Today at a glance

0:00	Check-in
0:02	Lesson 1 · Method → organization → product
0:12	Activity 1 · the innovation triangle
0:17	Lesson 2 · Team execution and the **3E's** (Empower, Experiment, Evolve)
0:27	Activity 2 · three sentences
0:32	Lesson 3 · Lean Startup and the Business Model Canvas
0:42	Activity 3 · fun-quiz ×4
0:47	Lesson 4 · Adaptation and system design
0:57	Activity 4 · flash-cards ×6 + zen-break
1:00	Team Studio · Lab 04

7 Where we are

1. Sprint 2 runs October 26 → November 8: today is its second week
2. Last week: your board became a measuring instrument (**WIP** (Work In Progress), lead time, cycle time)
3. **Today: the product in its business** — users, payers, value, and the cost of change
4. **Sunday, November 8, 23:59: Lab 04 feedback date** — your Mid-Term Basis Kit is complete
5. November 9: Mid-Term Review · November 16: Mid-Term

7 What you will be able to do after this hour

1. Explain why the way an organization works decides the product it delivers (Denning, Conway, **CI** (Continuous Integration), **CD** (Continuous Delivery), **DevOps** (Development and Operations))
2. State the **3E's** (Empower, Experiment, Evolve) and tell product discovery from product delivery
3. Write a hypothesis that can fail and fill the five customer-side blocks of a Business Model Canvas
4. Show how one design decision changes **WIP** (Work In Progress), cycle time and lead time

1

LESSON 1 · 10 MINUTES

Method → organization → product

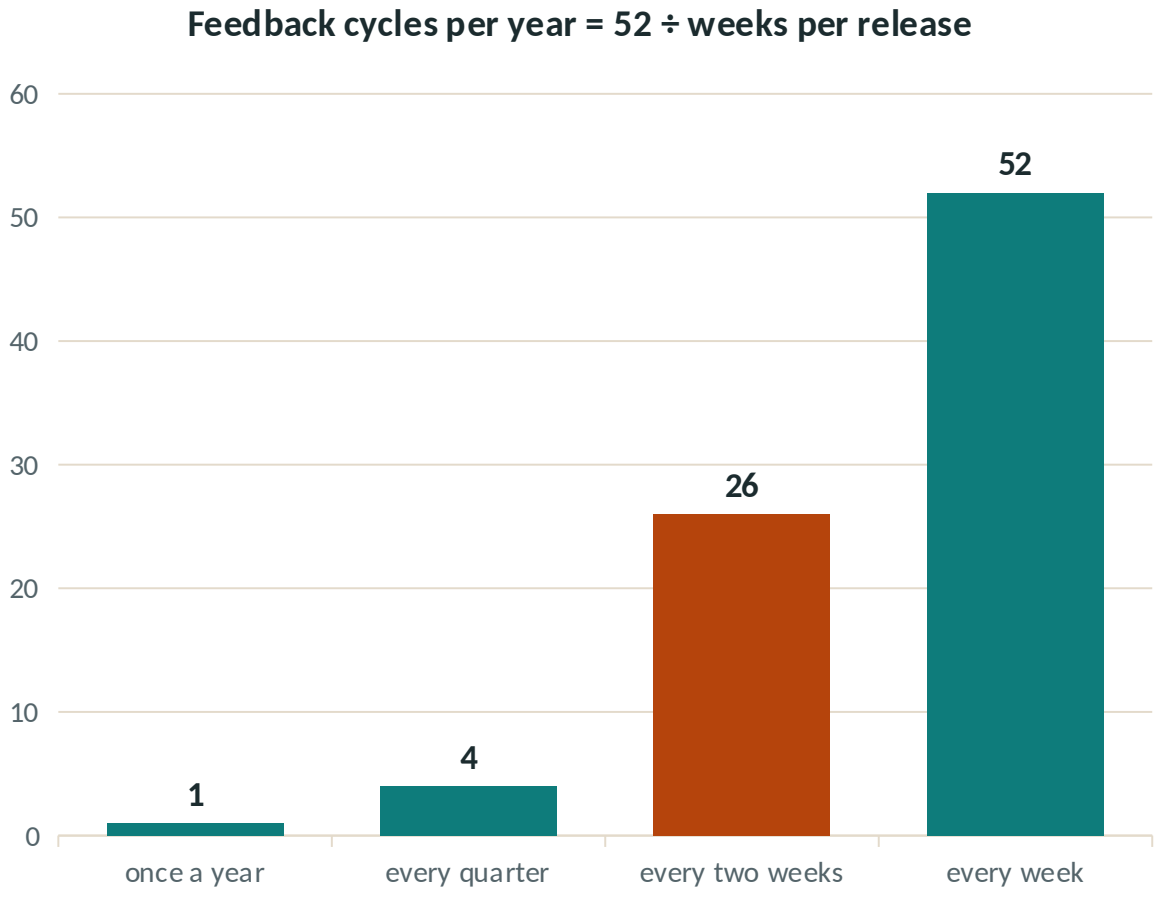
How the way of working decides what the customer receives

Denning's three laws · Conway's Law · Scrum's path to the digital workplace · the innovation triangle

L1

A method is a decision about the product

- Release once a year → learn from users at most once a year
- Usable increment every two weeks → up to 26 chances a year (52 ÷ 2)
- Same people, different product: the second team changed its product from real feedback 25 more times
- The method fixes: how often users receive something new · how soon you learn · how much a change of direction costs



L1

Denning's three laws

- **Law of the Small Team** — small, autonomous, cross-functional teams; short cycles; small tasks; continuous user feedback
- **Law of the Customer** — the purpose is value for customers; everyone sees how their work contributes
- **Law of the Network** — a network of teams toward a common goal, not a top-down hierarchy
- Source: Denning, Strategy & Leadership (2016); The Age of Agile (2018)

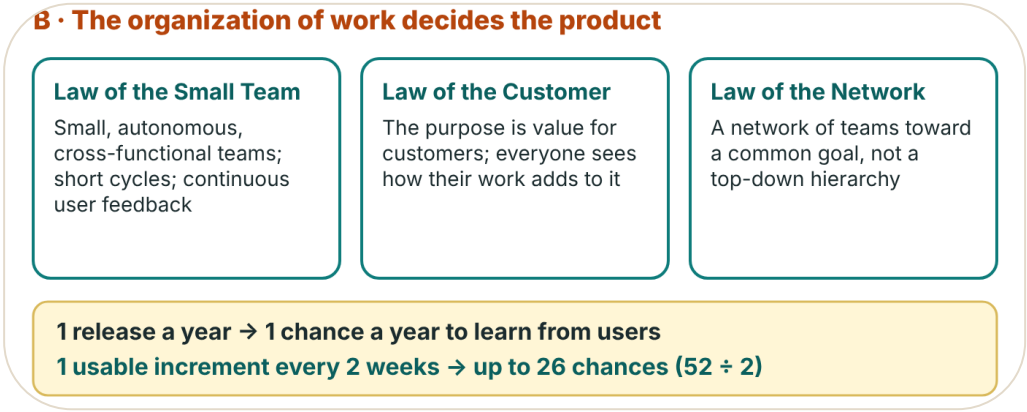


Diagram 1 · Denning's three laws

L1

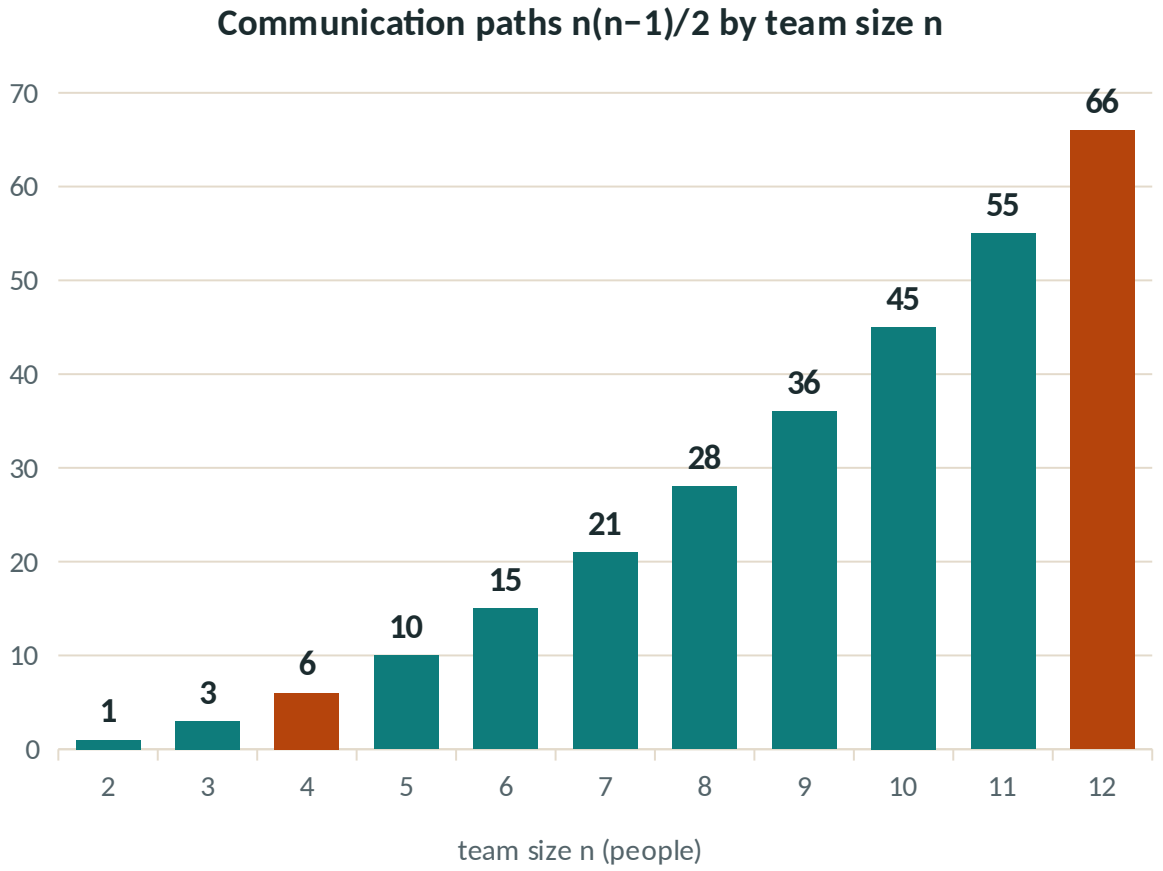
Two cases: Ericsson and Spotify

1. Ericsson before 2011: systems built on a five-year cycle, units of thousands of people
2. After the move to Agile: less work in progress, deployments one to two years earlier
3. Spotify: Discover Weekly built by one of more than 100 small teams
4. Barclays, a bank over three centuries old: Agile made a strategic initiative in 2015
5. **What changed in each: the organization of work** — and the product with it

L1

Conway's Law and the two-pizza team

- Conway (1968): organizations produce system designs that copy their communication structures
- Three departments that rarely talk → three parts that connect badly
- Amazon's two-pizza rule: teams under about ten people, each owning one product or service end to end
- When work grows: split the team, do not enlarge it
- Communication paths = n(n−1)/2 → 4 people: 6 · 12 people: 66



L1

Scrum's path: 1986 → 2020

- 1986 · Takeuchi & Nonaka, Harvard Business Review: overlapping phases, self-organizing teams
- 1995 · Schwaber & Sutherland first co-present Scrum at **OOPSLA** (Object-Oriented Programming, Systems, Languages and Applications conference)
- 2001 · Agile Manifesto · 2010 · first Scrum Guide
- 2020 · Scrum Guide: “even less prescriptive”, remaining IT-specific wording removed
- Scrum outside **IT** (Information Technology) is what the authors wrote, not a local variation

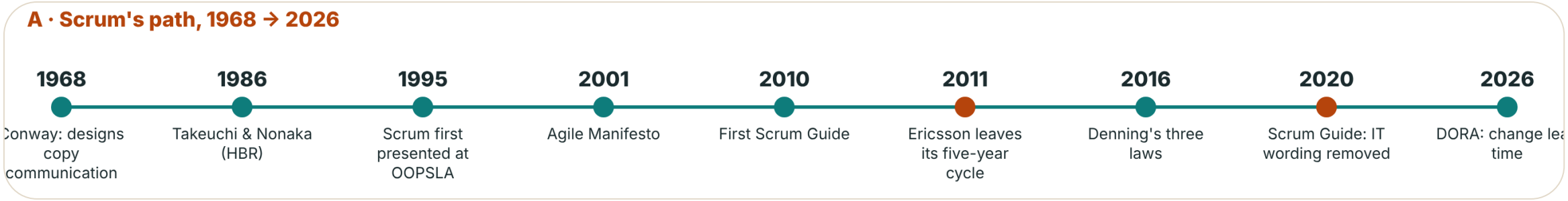


Diagram 1 · Scrum's path, 1968 → 2026

L1

The digital workplace: CI, CD, DevOps

- **CI (Continuous Integration)** — integrate every member's work at least daily; automated build and tests check each integration
- **CD (Continuous Delivery)** — changes of all types reach users quickly, safely, sustainably; the product is always releasable
- **DevOps (Development and Operations)** — builders and operators as one team, shared responsibility for the product in use
- None is required by the Scrum Guide; all exist because increments must reach users
- **DORA** (DevOps Research and Assessment)'s change lead time = committed to version control → deployed in production

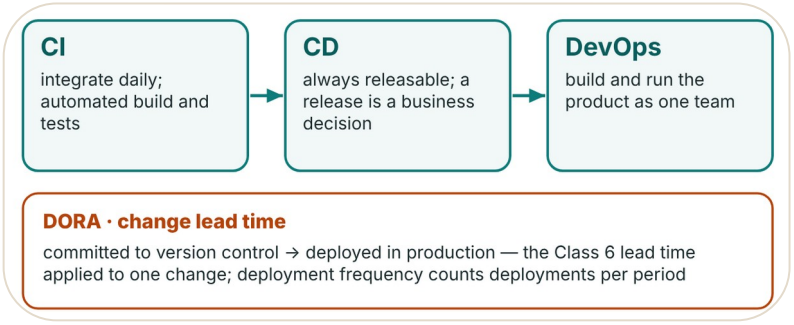


Diagram 1 · CI → CD → DevOps, and change lead time

L1

Product, service, and the value map

- Product = what the user receives (the allergen page)
- Service = the continuing activity that keeps value reaching the user (daily entry before 11:00, corrections within a day)
- A page nobody updates shows yesterday's allergens
- Value map: who builds · who uses · who pays or carries the cost · what flows along each arrow
- **An empty box is a finding: "unknown** — to ask whom, by when" (Lab 04, Box 3)

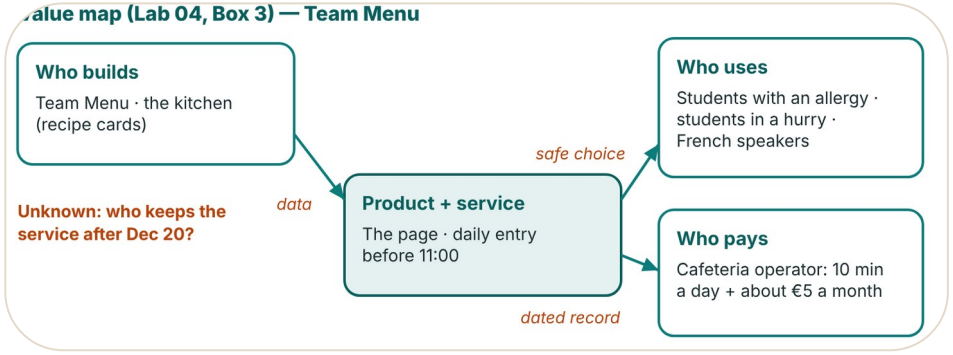


Diagram 1 · Team Menu's value map

L1

The innovation triangle

- **Desirability** — do people want it? → Design Thinking (Classes 10–13)
- **Viability** — does it sustain itself? → Lean Startup, the canvas (today)
- **Feasibility** — can we build and deliver it? → Scrum, Kanban (Classes 3–6)
- The corners touch: the triangle says which framework to reach for first

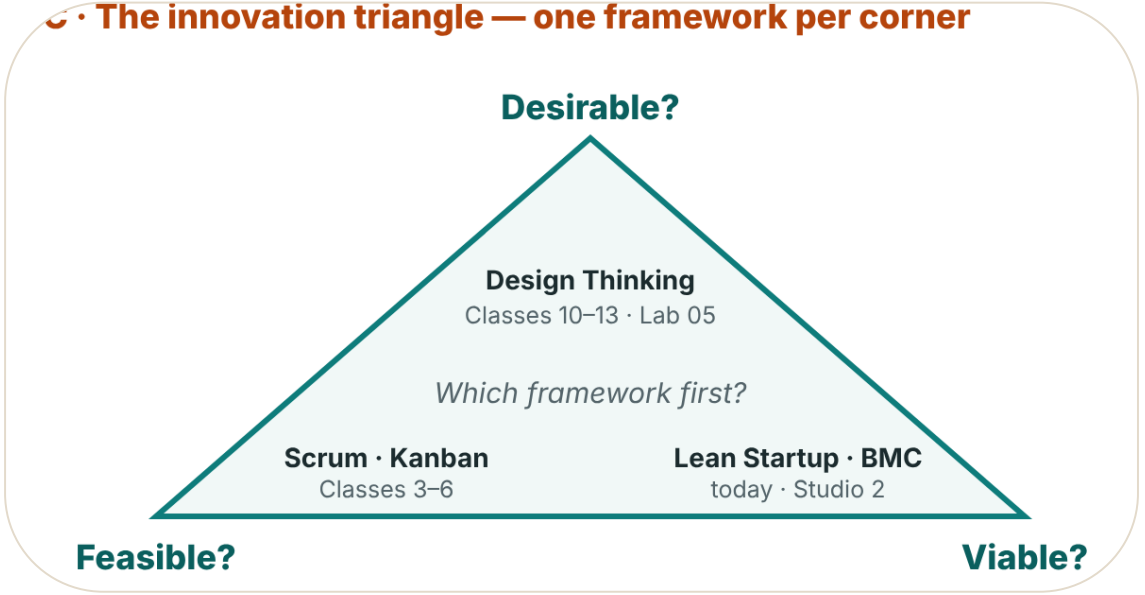


Diagram 1 · the innovation triangle

L1

Activity 1 · The innovation triangle

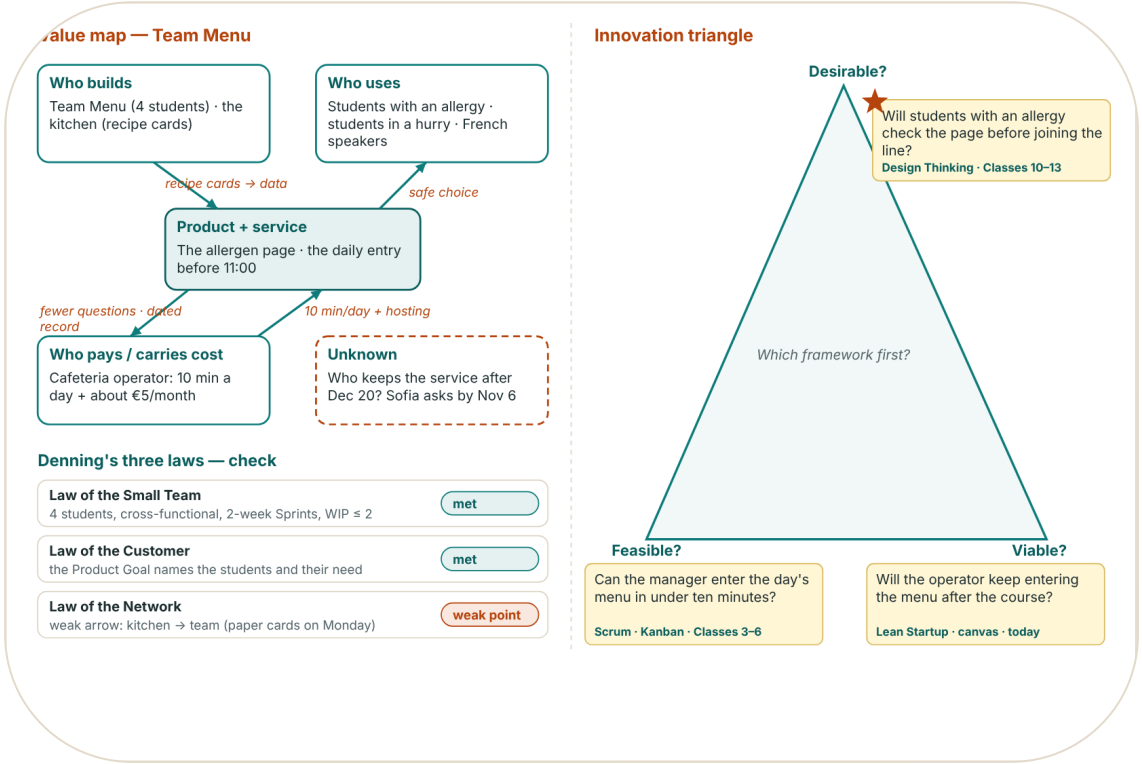
⏱ 5 min

1. **1 min** — draw the triangle; label Desirable? · Viable? · Feasible?
2. **2 min** — one open question about your product per corner (“will…?”, “can…?”, “who…?”)
3. **1 min** — star the question that would make the product pointless if the answer were “no”
4. **1 min** — two teams read their star; the class names the corner
5. Keep the photo: today's Step 3 starts from your star

L1

Team Menu's value map and triangle

- Who builds: Team Menu + the kitchen (recipe cards)
- Who uses: students with an allergy; students in a hurry; French speakers
- Who pays: the cafeteria operator (10 minutes a day + about €5/month hosting)
- Weak arrow: kitchen → team (cards on paper, Monday mornings)
- ★ Will students check the page before joining the line?



Code example 1 · value map and triangle (screenshot)

L1 Lesson 1 · Watch and read

Spotify Engineering Culture — Part 1 (Henrik Kniberg, 13:13)
https://www.youtube.com/watch?v=Yvfz4HGtoPc

CI/CD Explained — How DevOps Use Pipelines for Automation (Akamai Developers, 6:39)
https://www.youtube.com/watch?v=M4CXOocovZ4

Denning (2016), Understanding the three laws of Agile
https://www.emerald.com/insight/content/doi/10.1108/SL-09-2016-0074/full/html

Conway (1968), How Do Committees Invent?
https://www.melconway.com/Home/Committees_Paper.html

Scrum Guide 2020 and its revisions page · DORA delivery metrics
https://scrumguides.org/revisions.html

2

LESSON 2 · 10 MINUTES

Team execution and the 3E's

Empower, Experiment, Evolve — and discovery next to delivery

Two tests for every E sentence · Type 1 and Type 2 decisions · one board, two lanes

L2

Team execution: three conditions = the 3E's

- Team execution: turning a goal into a usable result, again and again, under uncertainty
- **Empower** — the team may decide how to reach its goal
- **Experiment** — the team tries ideas cheaply before committing
- **Evolve** — the team learns on a fixed rhythm
- The **3E's** (Empower, Experiment, Evolve) are this course's working framing, not a published framework

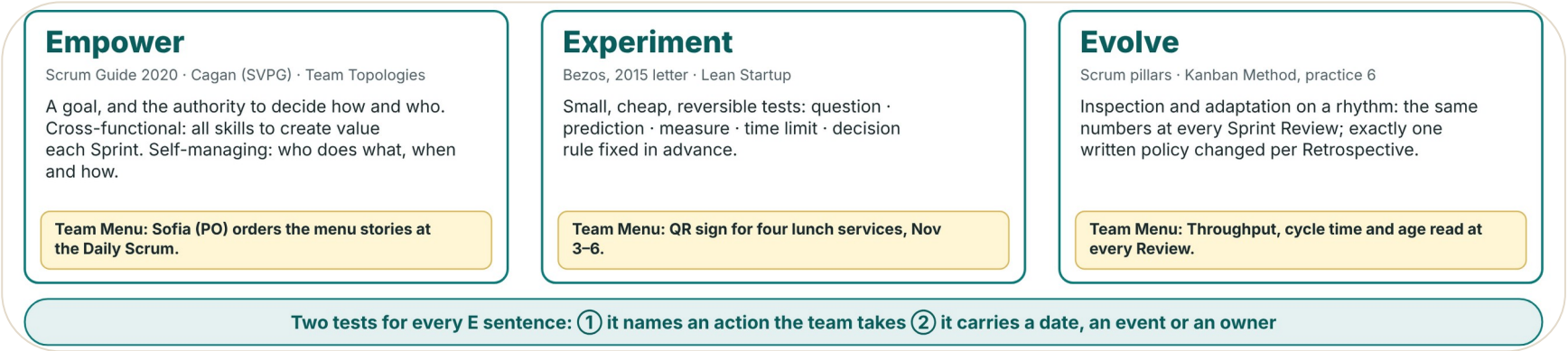


Diagram 2 · the 3E's with their sources and the two tests

L2

Two tests for an E sentence

1

Test 1

does it name an action the team takes (a decision, a test, a rhythm)?

2

Test 2

does it carry a date, an event or an owner?

3

× “We empower everyone.”

4

✓ “The **PO** (Product Owner) orders the backlog alone after hearing the team, at the Daily Scrum.”

L2

Empower — a goal, and the authority to reach it

1. Cross-functional: “the members have all the skills necessary to create value each Sprint”
2. Self-managing: “they internally decide who does what, when, and how” (Scrum Guide 2020)
3. Empowered team: a problem + accountability for the result · feature team: features already designed (Cagan)
4. Stream-aligned team: owns a flow of work end to end (Team Topologies)
5. Goals give the what; empowerment is the how and the who

L2

Experiment — Type 1 and Type 2 decisions

- Type 1: consequential and irreversible or nearly so → decide slowly and carefully
- Type 2: changeable and reversible → decide quickly, in small groups
- **Large organizations apply the Type 1 process to Type 2 decisions** — and slow down
- Source: Amazon, 2015 Letter to Shareholders

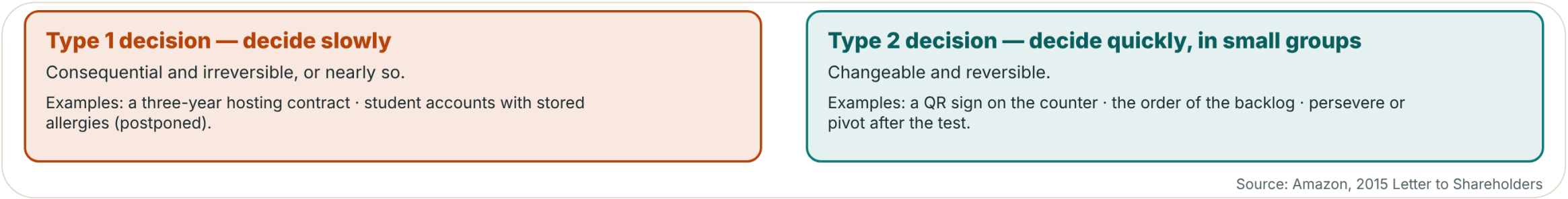


Diagram 2 · Type 1 and Type 2 decisions

L2

Experiment — the five parts

1. a question
2. a prediction written before the test
3. a measure that can be counted
4. a time limit
5. a decision rule fixed in advance

- A sign on a counter for four days instead of an app for a semester
- An experiment whose result cannot change a decision is not worth running

L2

Evolve — a rhythm, one policy per Retrospective

1. Scrum: inspection and adaptation · Kanban: “improve collaboratively, evolve experimentally”

2. Every Sprint Review: the same numbers

   velocity, throughput, cycle time, age of open cards

3. Every Retrospective: exactly one written policy changed, with an owner and a check date

4. Five changes at once: you cannot tell which one helped

L2

Discovery and delivery in one team

1. Discovery: “quickly generating validated product backlog items”
2. Delivery: “generating releasable software” (Cagan, 2012)
3. In parallel, by the same people

   now called continuous discovery and continuous delivery
4. Four questions of discovery: valuable? usable? feasible? viable?

L2

One board, two lanes

- Discovery cards (interview, **MVP** (Minimum Viable Product) test, spike, stakeholder question) next to delivery cards (stories)
- Course rule: discovery **WIP** (Work In Progress) ≤ 1
- The Sprint Review is where evidence changes the order of the backlog
- A new product needs more discovery than a mature one

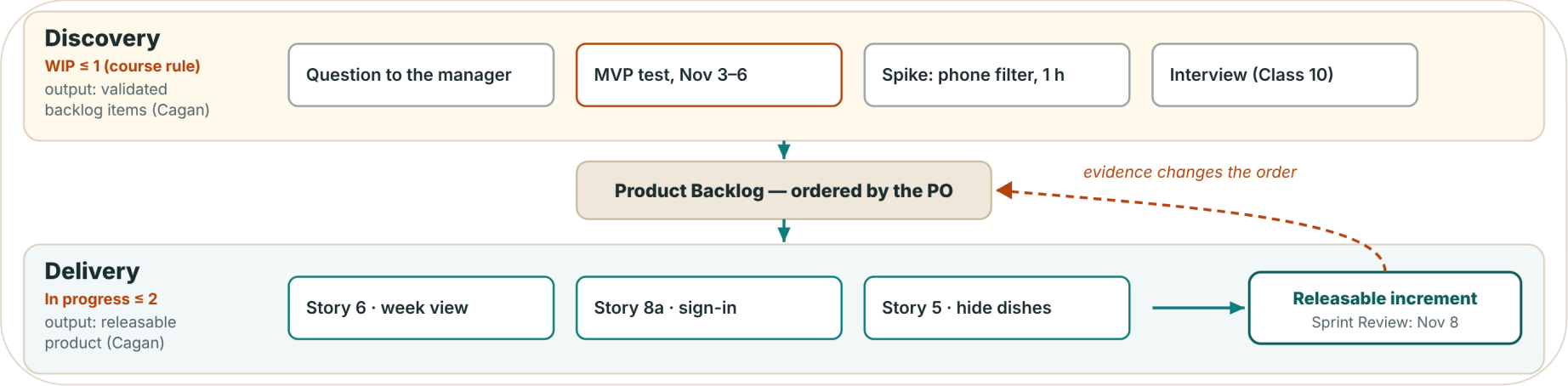


Diagram 2 · one board, two lanes

L2

Activity 2 · Three sentences

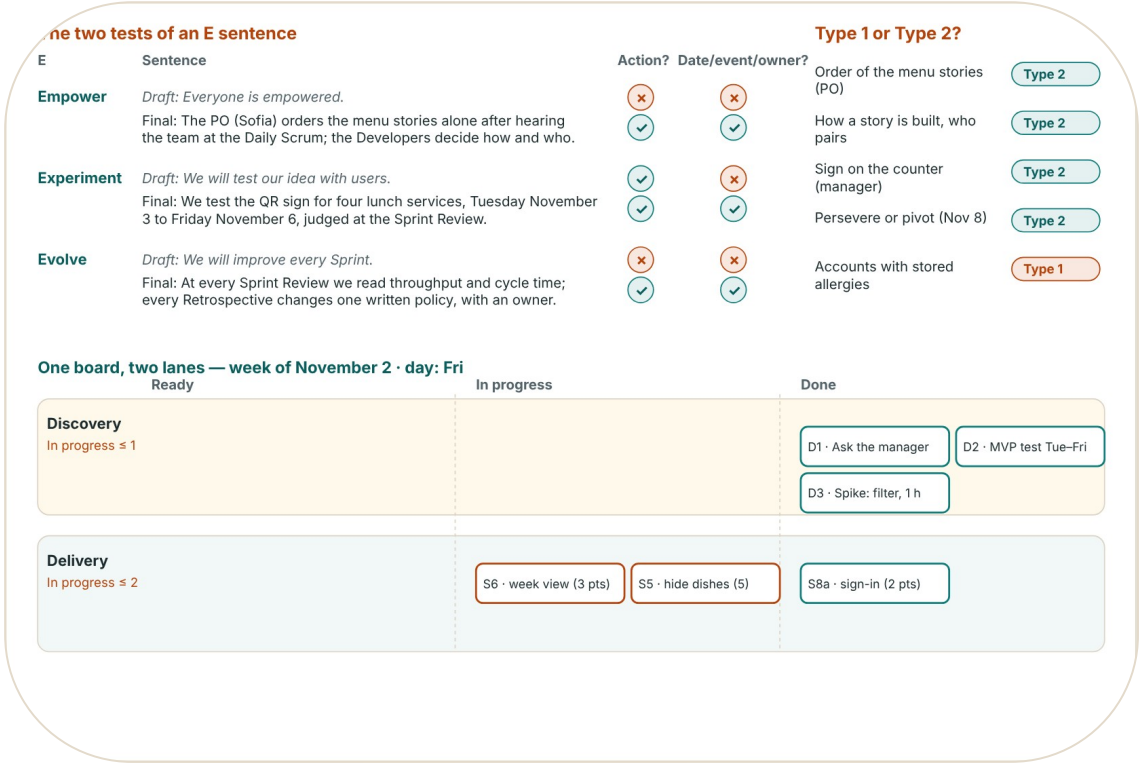
⏱️ 5 min

1. **3 min, alone** — one sentence per E for your product
2. Empower: who decides what, without asking whom
3. Experiment: which small, reversible test, judged when
4. Evolve: which numbers at which event; how many policies per Retrospective
5. **1 min, in pairs** — test each sentence twice (action? date / event / owner?)
6. **1 min** — two volunteers read their best sentence

L2

Team Menu's three E sentences

- Empower: Sofia (**PO**) orders the menu stories after hearing the team, at the Daily Scrum
- Experiment: **QR** (Quick Response) sign, four lunch services (Nov 3–6), judged at the Sprint Review on Nov 8
- Evolve: same numbers at every Review; one written policy per Retrospective
- Board: discovery ≤ 1 (manager question → **MVP** (Minimum Viable Product) test → spike) · delivery ≤ 2



Code example 2 · E-sentence checker and two-lane board (screenshot)

L2 Lesson 2 · Watch and read

Spotify Engineering Culture — Part 2 (Henrik Kniberg, 12:35)
https://www.youtube.com/watch?v=vOt4BbWLWQw

Four Questions to Ask Before You Design a Product — Marty Cagan (20VC, 2:21)
https://www.youtube.com/watch?v=cZWl5H-sz8M

Scrum Guide 2020 — cross-functional, self-managing
https://scrumguides.org/scrum-guide.html

Cagan (2012), Dual-Track Agile — SVPG
https://www.svpg.com/dual-track-agile/

Amazon, 2015 Letter to Shareholders — Type 1 / Type 2
https://s2.q4cdn.com/299287126/files/doc_financials/annual/2015-Letter-to-Shareholders.PDF

3

LESSON 3 · 10 MINUTES

Lean Startup and the Business Model Canvas

Build–Measure–Learn, a hypothesis that can fail, and the nine blocks

Riskiest assumption · MVP · decision rule · persevere or pivot · canvas v0

L3

Lean Startup: Build–Measure–Learn

- Eric Ries (2011): learning “can be validated scientifically, by running experiments”
- **Build** — the smallest product that can test the idea
- **Measure** — what real users do, not what they say
- **Learn** — persevere or pivot
- Unit of progress: validated learning, not features built

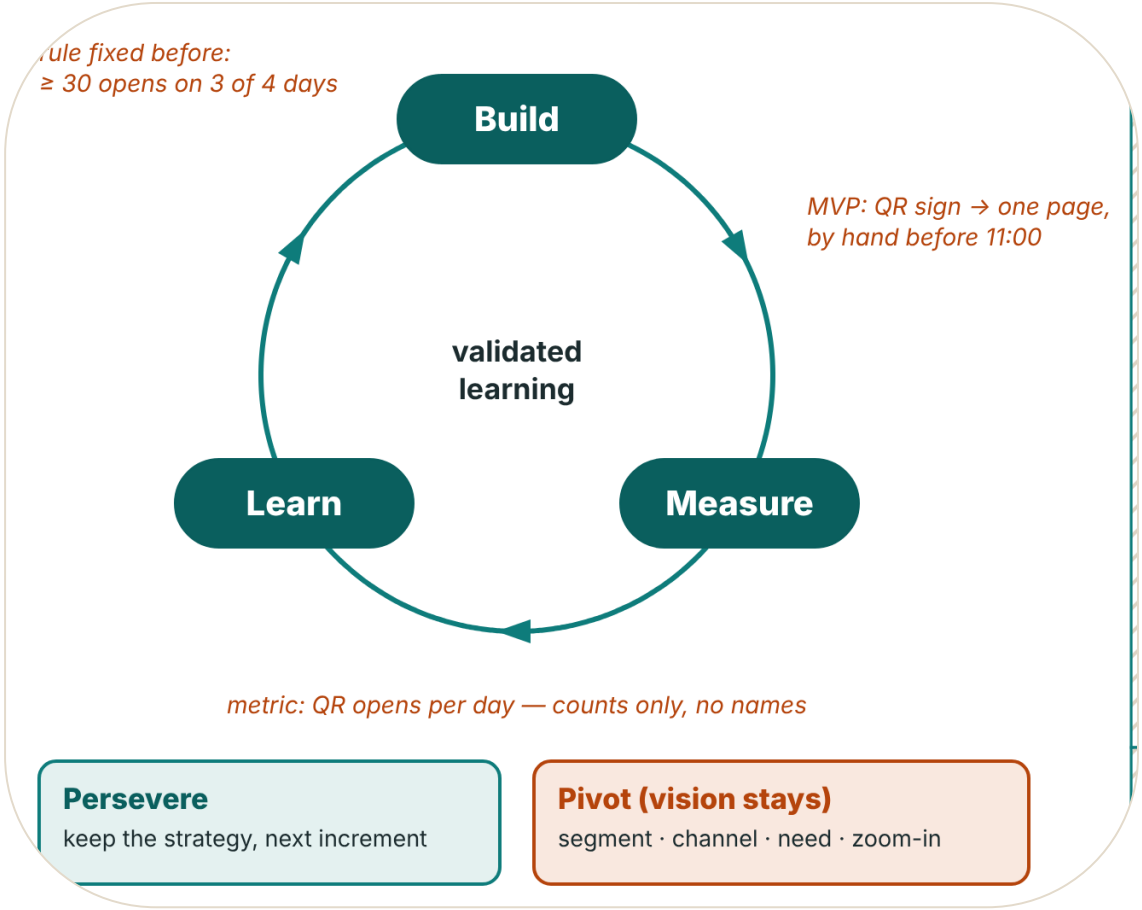


Diagram 3 · Build–Measure–Learn with Team Menu's MVP

L3

Lean Startup inside Scrum

1. The Sprint is the time-box

2. The **MVP** (Minimum Viable Product) test is a discovery card on the board

3. The Sprint Review is the Learn step, with stakeholders in the room

4. Scrum gives the rhythm; Lean Startup gives the question to answer in each rhythm

L3

Riskiest assumption and MVP

1. Riskiest assumption: false → the product is pointless, however well built
2. Usually about behaviour: “students will check a page” is riskier than “we can build a page”
3. **MVP** (Minimum Viable Product) (Ries): the version that collects the maximum validated learning with the least effort
4. Defined by the question it answers, not by its polish
5. Needs more than a day or two to build? Not minimal yet

L3

A hypothesis that can fail

We believe that [these people] will [do this] because [reason].

We will know we are right when we see [metric] reach [number] by [date].

- Six parts: segment · behaviour · reason · metric · number · date
- No number and no date → it cannot fail → it cannot teach
- Template extended from Gothelf & Seiden, Lean **UX** (User Experience)

L3

Metric and decision rule

- Count without names: opens, sign-ups, returning visitors, minutes saved
- Actionable metric: shows cause and effect, supports a decision
- Vanity metric: grows and looks good, supports no decision (“total visits since September”)
- Decision rule written before the test: the number for “persevere”, the pivot otherwise

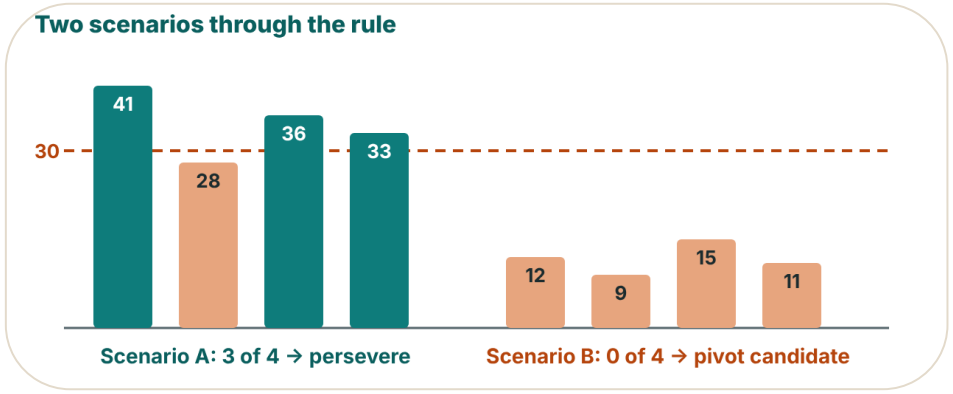


Diagram 3 · two scenarios through the rule

L3

Persevere or pivot

1

Persevere: the hypothesis held

keep the strategy, build the next increment

2

Pivot: a structured change of strategy to test a new fundamental hypothesis; the vision stays

3

Types (Ries): customer segment · channel · customer need · zoom-in

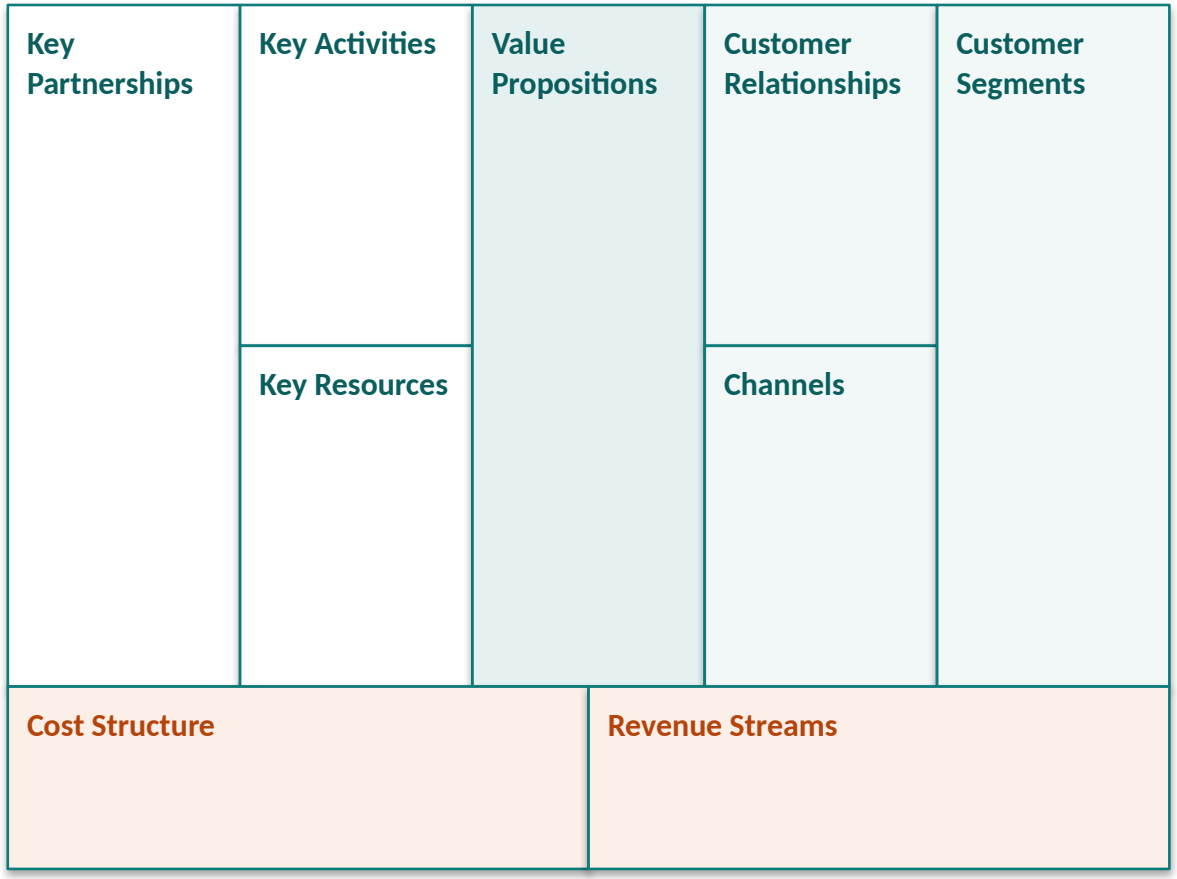
4

Decided by the whole team at the Sprint Review, with the rule fixed before

L3

The Business Model Canvas — nine blocks

- Right · customers and value: Customer Segments · Value Propositions · Channels · Customer Relationships
- Left · infrastructure: Key Activities · Key Resources · Key Partnerships
- Bottom · money: Cost Structure · Revenue Streams
- Value Proposition in the centre: every block creates, delivers or captures it
- Value Proposition Canvas: jobs, pains, gains of one segment (Strategyzer)



Osterwalder & Pigneur (2010): customers and value right · infrastructure left · money at the bottom

L3

Canvas v0 and v1 — two rules

1. v0 today: Customer Segments, Value Propositions, Channels, Customer Relationships, Revenue Streams
2. v1 in Studio 2 (December 7): all nine blocks
3. Rule 1: every segment is a **D2** (Deliverable 2, Product Vision Board) user or a payer named by role
4. Rule 2: every value proposition states a gain, not a feature
5. The hypothesis tests one line through the canvas: segment → value proposition → channel

L3

Activity 3 · Fun-quiz ×4

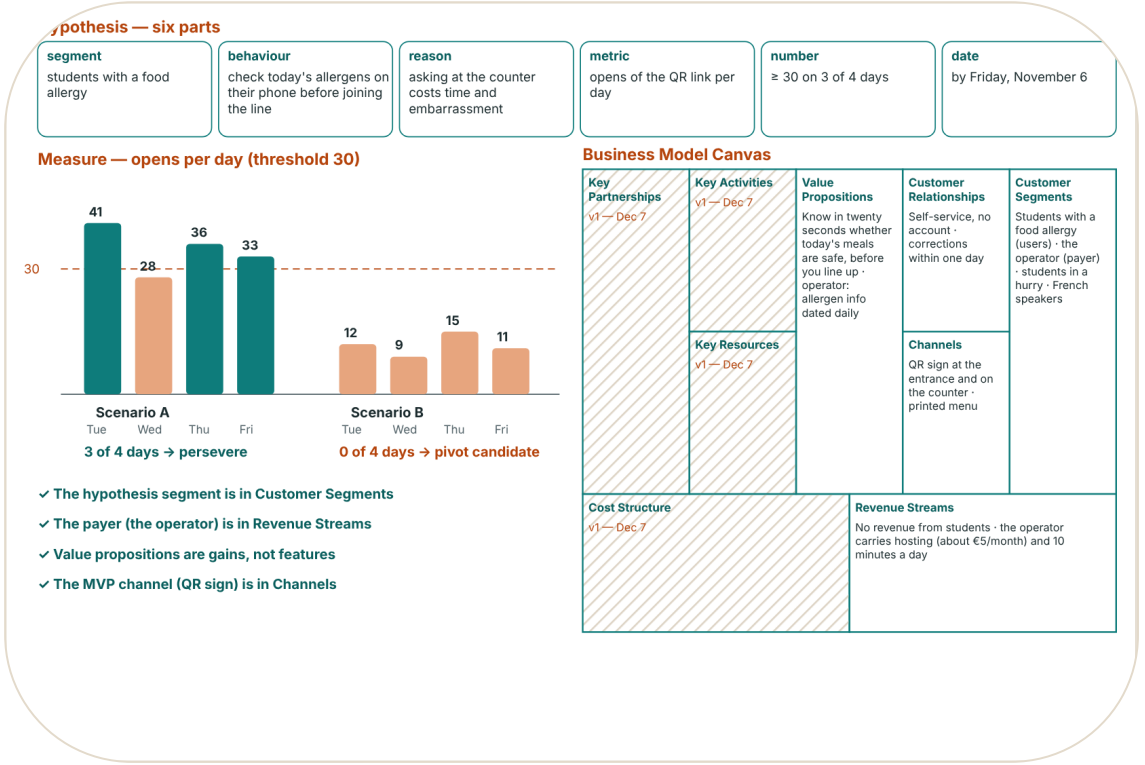
⏱️ 5 min

1. 1 · Which block holds “students who commute”?
2. 2 · Which block holds “subscription, 3 $ per month”?
3. 3 · In Build–Measure–Learn, what comes right after the **MVP** (Minimum Viable Product)?
4. **4 · A pivot changes what** — and keeps what?

L3

Team Menu's MVP test — two scenarios

- **MVP** (Minimum Viable Product): **QR** (Quick Response) sign at the entrance and on the counter → one page, today's three hot meals, by hand before 11:00
- Not the filter, not French, not the week view · Tue Nov 3 – Fri Nov 6
- Rule: ≥ 30 opens on at least 3 of 4 days
- Scenario A: 41 · 28 · 36 · 33 → 3 days → persevere
- Scenario B: 12 · 9 · 15 · 11 → 0 days → pivot candidate (channel or segment)



Code example 3 · MVP rule and canvas v0 (screenshot)

L3

Team Menu's canvas v0

- Segments: students with an allergy (users) · the operator (payer) · secondary groups
- **Value: know in twenty seconds whether today's meals are safe** — before lining up
- Channels: **QR** (Quick Response) sign · printed menu · later the student app
- Relationships: self-service; corrections within a day; monthly check-in
- Revenue: none from students; the operator carries hosting and 10 minutes a day

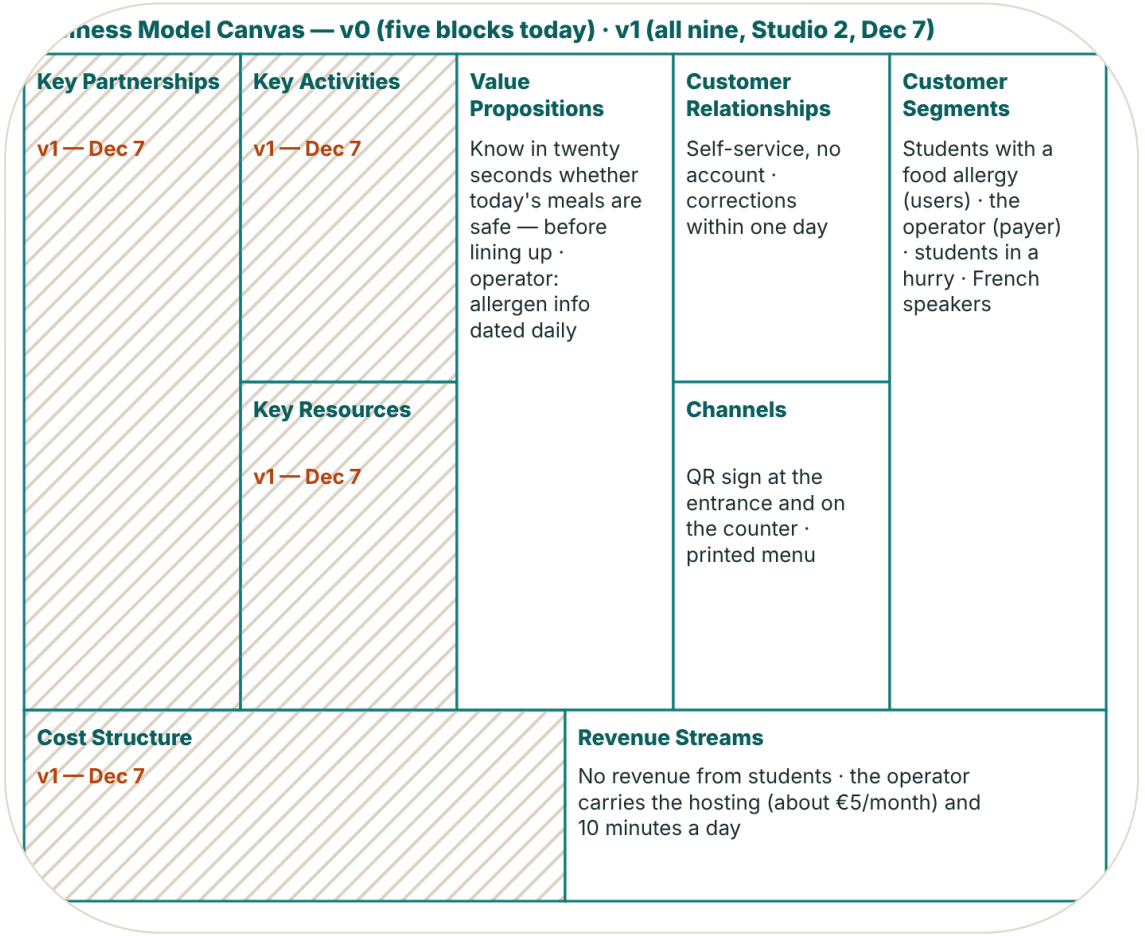


Diagram 3 · Team Menu's canvas v0 (v1 blocks hatched)

L3 Lesson 3 · Watch and read

Validate your business idea: The Lean Startup (Productivity Game, 8:48)
https://www.youtube.com/watch?v=QaoVWtLX038

Business Model Canvas Explained (Strategyzer, ≈ 2 min)
https://www.youtube.com/watch?v=QoAOzMTLP5s

Strategyzer's Value Proposition Canvas Explained (Strategyzer, 3:13)
https://www.youtube.com/watch?v=ReM1uqmVfP0

The Lean Startup — principles (Eric Ries)
https://theleanstartup.com/principles

Strategyzer — the official Business Model Canvas
https://www.strategyzer.com/library/the-business-model-canvas

4

LESSON 4 · 10 MINUTES

Adaptation and system design

Technical debt, components and interfaces — and what a design does to your lead times

Change without chaos · information hiding · API-first · Monolith First · Little's Law again

L4

Change without chaos

1. Principle 2: “Welcome changing requirements, even late in development”
2. Principle 9: “Continuous attention to technical excellence and good design enhances agility”
3. 1 · change enters through one door: the Product Backlog
4. 2 · the Sprint Goal stays stable
5. 3 · the design keeps the cost of a typical change small

L4

Technical debt

1. Ward Cunningham, **OOPSLA** (Object-Oriented Programming, Systems, Languages and Applications conference) 1992 (the WyCash report)
2. Technical debt: the extra work every future change costs because the code reflects an earlier understanding or a shortcut
3. Interest: the extra work on each later change
4. Refactoring: rewriting the design to match the current understanding
5. Never meant as permission to write careless code

L4

Four kinds of technical debt (Fowler, 2009)

- **Prudent · deliberate** — “ship now, correct it after the test”
- **Reckless · deliberate** — “no time for tests”
- **Prudent · inadvertent** — “now we see how it should have been designed”
- **Reckless · inadvertent** — design practice unknown, cost unnoticed
- Team Menu: names typed into the page → prudent, deliberate; corrected October 30

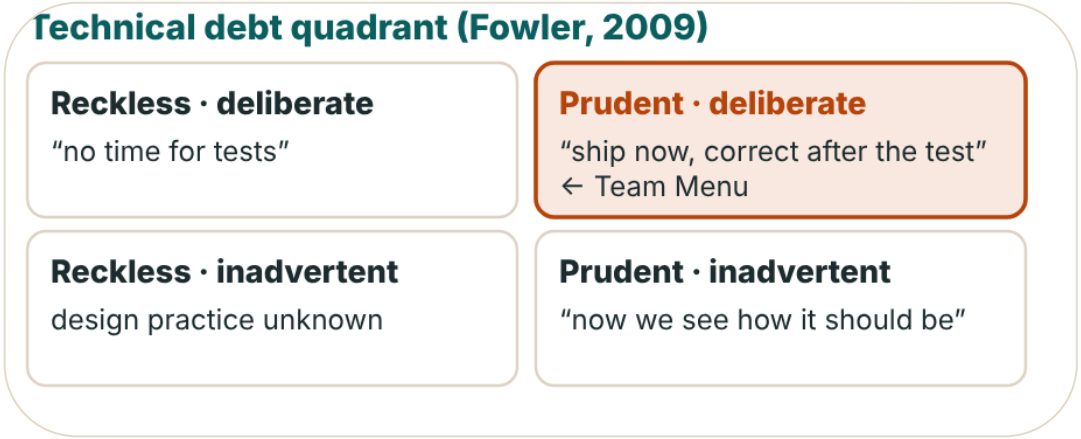


Diagram 4 · Fowler's technical debt quadrant

L4

Components, interfaces, information hiding

- Component: a named part with one responsibility
- Interface: the agreed way it exchanges information with the others
- Parnas (1972): each module hides one design decision likely to change
- When that decision changes, only that module changes
- Cannot name your components? You cannot tell which one a request touches

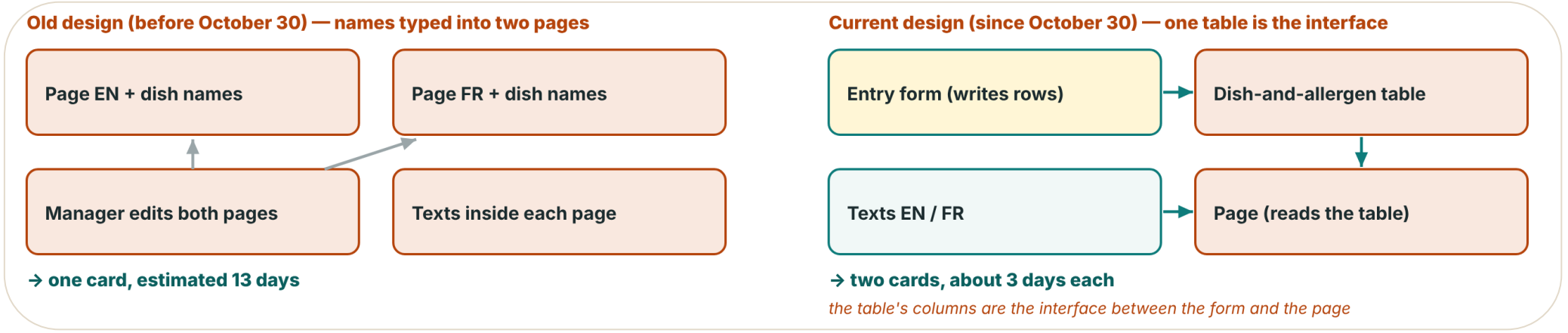


Diagram 4 · old and current design of Team Menu's product

L4

APIs, API-first, Monolith First

1. **API** (Application Programming Interface): the requests a component accepts and the responses it returns

2. API-first: agree the interface before the implementation (OpenAPI) — both sides built and changed independently

3. Monolith: one deployable unit · microservices: many independently deployable services

4. Monolith First (Fowler, 2015): one well-structured unit, split when boundaries are clear

5. Loosely coupled architecture: small teams deploy safely and independently (**DORA**)

L4

How a design decision changes WIP and lead times (1/2)

Coupled design → a typical change touches many components

↓

one large card → long cycle time

↓

people start other cards while it waits → **WIP** (Work In Progress) rises

↓

Little's Law: average cycle time = average **WIP** ÷ throughput → longer for all

The other design

Loosely coupled → one or two small cards → under the limit → out in days

L4

How a design decision changes WIP and lead times (2/2)

- Request: “cover the small café on the top floor as well”
- Old design: one 13-day card → café usable day 13 · average cycle 7.7 d · lead 9.0 d
- Current design: two 3-day cards → café usable day 3 · average cycle 4.0 d · lead 5.5 d
- **Work needed: 23 against 16 pair-days** — 7 pair-days of interest
- Little's Law holds exactly: each window starts and ends empty



Diagram 4 · the café request under two designs

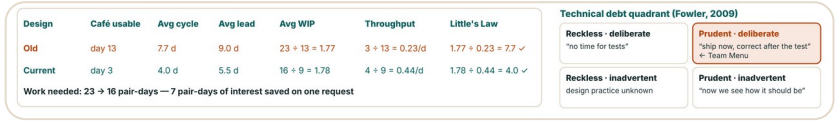


Diagram 4 · the numbers, with the Little's Law check

L4

Activity 4 · Flash-cards ×6 + zen-break

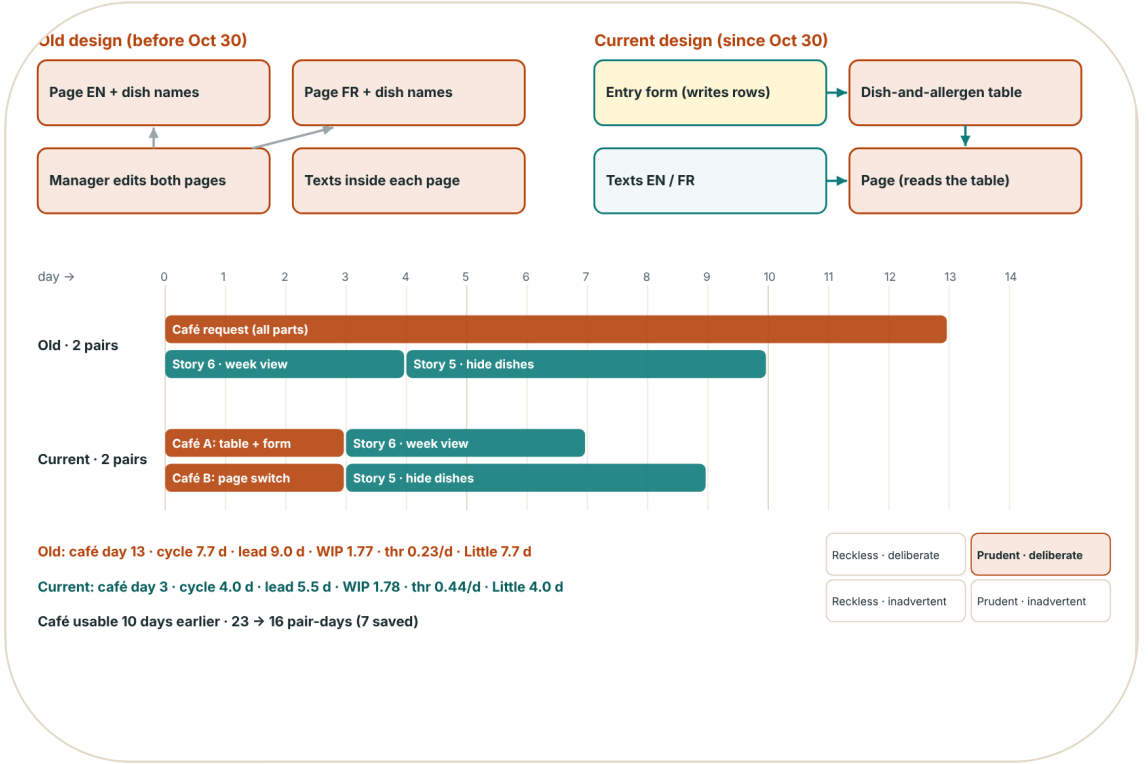
⏱️ 3 min

1. **MVP** (Minimum Viable Product) · pivot · value proposition
2. technical debt · product discovery · product delivery
3. Round 1: term → definition · Round 2: definition → term
4. **Then 60 seconds of stillness** — no phones, no talking

L4

Team Menu's adaptation line (Box 6)

- Components: the page · the dish-and-allergen table · the entry form · the texts
- The table's columns are the interface between the form and the page
- What if: the small café → table + page change; form slightly; texts untouched
- Effect: two cards of about three days, within our limit of two
- Debt: prudent and deliberate, corrected October 30



Code example 4 · adaptation and flow simulator (screenshot)

L4 Lesson 4 · Watch and read

Debt Metaphor (Ward Cunningham, ≈ 5 min)
https://www.youtube.com/watch?v=pqeJFYwnkjE

What's the Difference Between APIs, Services and Microservices? (IBM Technology, 12:18)
https://www.youtube.com/watch?v=qGFRbOq4fmQ

Fowler (2009), Technical Debt Quadrant
https://martinfowler.com/bliki/TechnicalDebtQuadrant.html

Parnas (1972), On the Criteria to Be Used in Decomposing Systems into Modules
https://doi.org/10.1145/361598.361623

Fowler (2015), Monolith First · DORA, loosely coupled teams
https://martinfowler.com/bliki/MonolithFirst.html

7 After the hour · Team Studio = Lab 04, Part A

10′	**Step 0 · Planning** — goal, four hats
30′	**Step 1 · Sprint 1 flow numbers** — Box 1 and Box 2 (Class 6)
20′	**Step 2 · value map + three E sentences** — Box 3 (Lessons 1–2)
25′	**Step 3 · assumption, MVP (Minimum Viable Product), hypothesis, metric** — Box 4 (Lesson 3)
20′	**Step 4 · canvas v0** — Box 5 (Lesson 3)
10′	Sprint Review · 5′ Retrospective and week plan

7 Home Sprint and what comes next

1. Home: Box 6 adaptation line · Box 7 Sprint 2 Review and Retro · checks · **PDF** (Portable Document Format) + Learning Log
2. **MVP (Minimum Viable Product) test Nov 3–6** — counts only, never names
3. Read: Denning, Chapter 7 “Changing the Organizational Culture”
4. **Sun Nov 8, 23:59** — Lab 04 feedback date · Mon Nov 9 — Mid-Term Review
5. **Mon Nov 16** — Mid-Term (Parts C and D use today's shapes)
6. **Mon Nov 23** — Lab 05 opens the Design Thinking Sprint (summary only)

7 Key links and the Legal Notice

Scrum Guide 2020 · Kanban Guide · The Lean Startup principles
https://scrumguides.org/scrum-guide.html

Strategyzer — Business Model Canvas · DORA — delivery metrics
https://www.strategyzer.com/library/the-business-model-canvas

Agile Manifesto principles · Fowler — Monolith First, Technical Debt Quadrant
https://agilemanifesto.org/principles.html

Check every link first: VirusTotal · Google Safe Browsing
https://www.virustotal.com/gui/home/url

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https://cyber-soho.github.io/prevmon/00.01.%20cyber.soho%20-%20legal.html